

M1 · IDENTITY CORE · ONE PILOT STORY

One identity. New Cardano authority.

What a team with a production KERI identity can build the day the M1 identity core lands.

• WIRE MERGED + BYTE-TESTED

• VALIDATOR + ENFORCEMENT SHIP WITH M1

They already have an identity. They need it to mean something on Cardano.

“As Northstar’s identity team, we want our existing KERI authority to control Cardano records, so that partners can rely on us without learning a second identity system.”

ALREADY OPERATES

A standard KERI AID

E... · Blake3 · Veridian-compatible

A 2-of-3 board signs today. Successor keys are committed and held separately.

NEEDS TO BUILD

An owned Cardano record

Counterparties must resolve current authority after rotations, using a stable identifier.

identity → authority → data

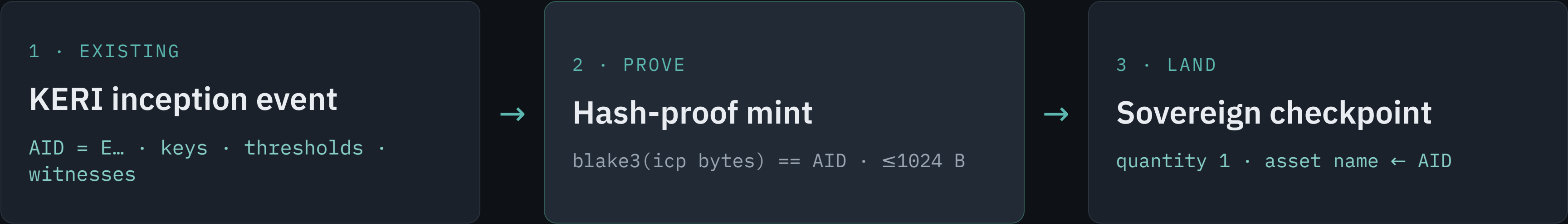
WILL NOT ACCEPT

A replacement identity

No re-issuance. No fixed hot key. No bridge custodian or oracle in the day-to-day authorization path.

Bring the identity you already operate.

“As the holder of a standard KERI AID, we can register it unchanged, so that the identity our partners already know gains a Cardano presence.”



HONEST BOUNDARY The hash proof binds event bytes to the AID. CESR → datum semantic projection is separately attested at registration and challengeable.

E-native no re-issuance or Cardano-specific AID

stable token deterministic asset name, quantity one

≤1024 B trustless byte binding for observed registering shapes

A stable name for changing authority.

“As a dApp author, I can bind to Northstar once, so that its contracts keep working after every legitimate key rotation.”

NEVER CHANGES

Checkpoint asset

policy_id
+
aid_asset_name

quantity = 1

THE DATUM ADVANCES

current keys	raw Ed25519 keys · zero-hash authorization	k / kt
next pair	explicit commitment · byte-equal to public KEL	n / nt
receipts	witness keys + threshold, carried explicitly	b / bt
sequence	Cardano projection + native KERI event position	seq / s

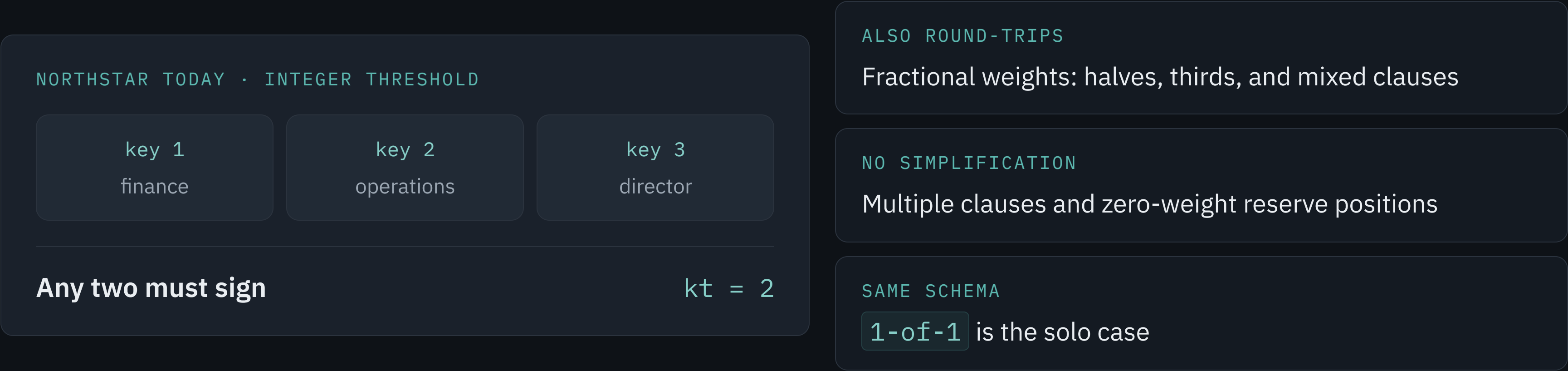
one lookup resolve by (policy, asset)

one shape versioned, byte-tested datum

same handle before and after rotation

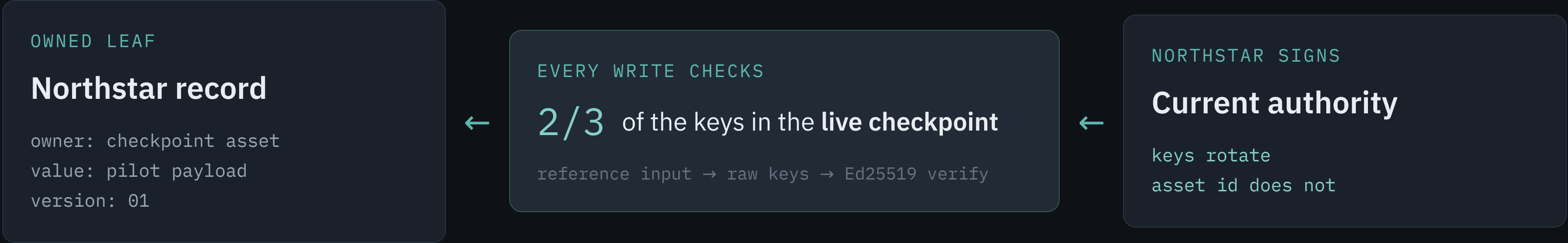
The board’s real rules survive the move.

“As a board of key holders, we can keep our actual signing policy, so that on-chain authority does not weaken governance for implementation convenience.”



Write once. Keep control after every rotation.

“As Northstar, we can own an on-chain record under our live identity, so that the record follows our authority rather than a fixed key.”



After rotation: same record, same owner handle, new signing keys.

ZERO-HASH AUTH

owned leaf writes authorize against current checkpoint stable owner record does not need migration on rotation composition shipped MPFS cage + M1 identity core

Rotate without asking anyone.

“As Northstar’s identity team, we can replace our operational keys ourselves, so that key hygiene never depends on an admin, allowlist, or oracle.”

KERI DUAL-THRESHOLD ADVANCE

commitment

revealed keys match the earlier `n` list

+

old nt

reveals satisfy the committed next threshold

+

new kt

signers satisfy the new current threshold

Checkpoint advances `seq + 1`

NO GATEKEEPER

REAL RESERVE PATTERN

Reveal 3 of 7

GLEIF Root’s production shape: restate the threshold and carry unexposed reserves forward.

3.6%

CPU / reveal

4.5%

mem / reveal

permissionless controller submits its own valid advance

partial reveal reserve rotations are representable

measured one Blake3 block per revealing key

Stolen current keys do not control the future.

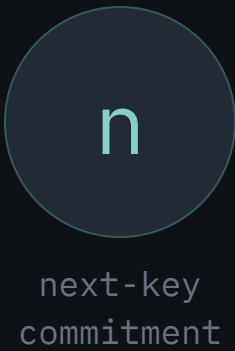
“As an identity owner whose current quorum was stolen, we can rotate the attacker out, so that compromise is an incident —not automatic identity death.”

ATTACKER

Has every current key

current quorum = satisfied

✗ rotation rejected



NORTHSTAR

Reveals committed next keys

$\text{blake3}(\text{qb64}(\text{key})) \in \text{prior } n$

✓ identity advances

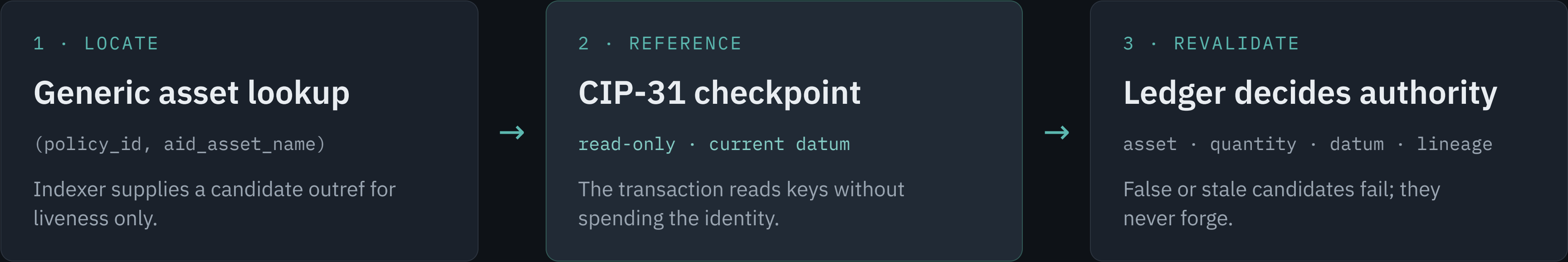
negative test full stolen current quorum cannot rotate

same AID successor keys evict the compromised set

hard boundary loss of current + next material is unrecoverable in M1

Contracts follow the identity, not its keys.

“As a contract author, I can resolve Northstar’s live authority in one reference input, so that my protocol survives rotations and fails closed on a freeze.”



Frozen means frozen: the accepted script address changes, so consumers fail closed structurally.

LIVENESS ≠ AUTHORITY

one lookup no KEL replay in the hot path stale indexer can delay, never forge rotation-safe same asset resolves the new keys

Lying has an on-chain consequence.

“As Northstar’s counterparty, I can rely on the checkpoint because equivocation is punishable by anyone, so that trust rests on evidence—not operator discretion.”

FORK

Signed alternative rotation

Burn

Permanent tombstone. Identity cannot be re-minted. Prover is paid from the deposit.

CARDANO BEHIND

Witnessed KERI rotation exists

Freeze

Advance-only until the owner catches up.
Consumers fail closed.

CARDANO AHEAD

Absence is not provable

Prevent

Witness receipts gate advance; later KEL movement yields convergence or conviction.

No framing: witness signatures alone cannot convict. Only the owner’s pre-committed signing threshold can.

SHIPS IN M1 VALIDATOR

The public evidence creates three useful roles.

The identity is the protagonist. Watchers, auditors, and issuers make its state enforceable, replayable, and consumable.

WATCHER

Compare. Prove. Get paid.

Diff public KEL fields against public checkpoints. Submit convict or freeze evidence.

permissionless
byte equality
bounty-funded

AUDITOR

Replay, don't trust.

Rebuild the KEL with dual-root reconstruction and cite chain evidence.

KEL replay
byte-for-byte
tombstones persist

ISSUER

Publish live status.

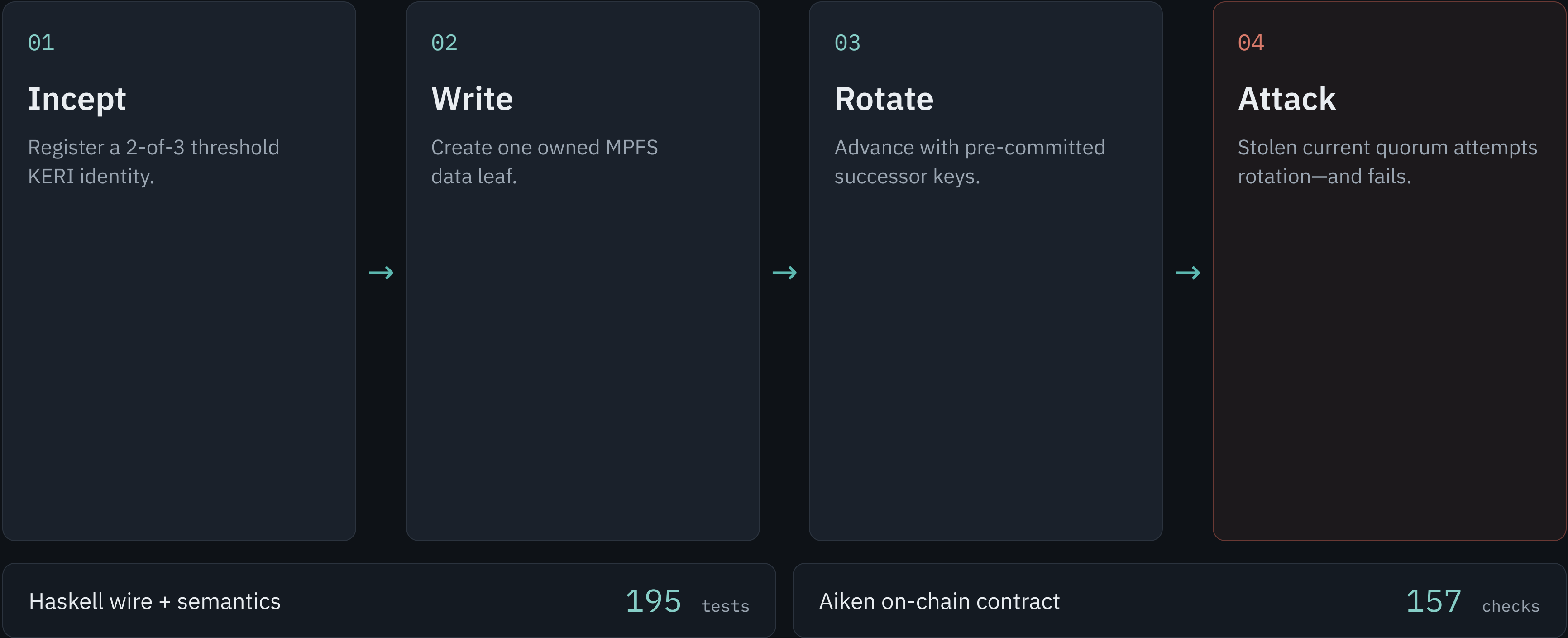
Mirror a per-issuer TEL revocation registry with ledger ordering.

32-byte IDs
replayable status
no issuer API hot path

SCOPE M1 mirrors credential status. Full GLEIF → QVI → LE credential-chain verification belongs to M2.

One terminal cast is the acceptance test.

If the identity cannot complete its journey end to end, the milestone is not done.



What M1 proves—and where it stops.

M1 ESTABLISHES

- ✓ Existing independent E-native AIDs register without re-issuance.
- ✓ Stable checkpoint authority survives real KERI rotations.
- ✓ Owned data, replayable evidence, lifecycle, and TEL status compose on top.
- ✓ Provable divergence freezes or burns without a committee.

NOT IN M1

- M2 Full GLEIF → QVI → LE credential-chain verification.
- M2 General authorization envelopes for arbitrary dApp actions.
- M3 KERI-wallet detached-signature UX for Veridian/Signify.
- M5 Delegated AIDs and superseding recovery.

A GOOD M1 PILOT BRINGS

one independent KERI AID · one Cardano record it should own · one explicit freshness policy

let's test the core →